NNN-T-780B
March 10, 1970
SUPERSEDING
Int. Fed. Spec. NNN-T-00780A(GSA-FSS)
June 7, 1968 and
Fed. Spec. NNN-T-780
March 13, 1963

FEDERAL SPECIFICATION

TUBE, CENTRIFUGE, GLASS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the various glass centrifuge tubes used in centrifuges in laboratory procedures.

1.2 Classification.

1.2.1 Types, capacities, and shapes. This specification covers one grade of centrifuge tubes of the following types, capacities, and shapes.

Type I - Graduated.

Capacity.

Milliliters:

6.5 - round bottom.
15.0 - conical bottom.
50.0 - conical bottom.
100.0 - (see 3.2.2.1.3).

Type II - Ungraduated.

Capacity.

Milliliters:

0.5 - conical bottom.
1.2 - conical bottom.
2.0 - conical bottom.
3.0 - conical bottom.
5.0 - conical bottom.
12.0 - conical bottom.
15.0 - conical bottom.
40.0 - conical bottom.
50.0 - conical bottom.
50.0 - conical bottom, screw-cap.
50.0 - round bottom.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

DD-G-541 - Glass (Laboratory).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-T-45 - Tape, Gummed Paper, Reinforced and Plain, for Sealing and Securing.

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 626/401 - Glassware, Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington

(Federal Government activities may obtain copies of Federal specifications, standards, and handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids shall apply.

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

American Society for Testing and Materials (ASTM) Standards:

D 91 - Precipitation Number of Lubricating Oils.
D 96 - Test for Water and Sediment in Crude Oils.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material. Centrifuge tubes shall be of borosilicate glass conforming to type I, class A of DD-G-541.

3.2 Design.

3.2.1 General. Centrifuge tubes shall be cylindrical in shape with a conical or round bottom (see 3.2.2.1). The top or open end shall be finished with a beaded rim; except for those where a pourout lip is specified and also the type II tubes capacities 0.5 ml. and 1.0 ml.

3.2.2 Type I.

3.2.2.1 Shape.

3.2.2.1.1 Capacity 6.5 milliliters (ml.). The 6.5 ml. capacity centrifuge tube shall be tapered to a smaller diameter cylindrical stem with a round bottom.

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3.2.2.1.2 Capacity 15 and 50 ml. The 15 and 50 ml. capacity centrifuge tubes shall have a conical bottom. The conical tip shall be convex and thickened.

3.2.2.1.3 Capacity 100 ml. The 100 ml. capacity centrifuge tube shall conform to ASTM D 91.

3.2.2.2 Dimensions. Dimensions of type I centrifuge tubes shall be as shown in table I.

TABLE I. Dimensions of type I centrifuge tubes

Capacity (ml.)	Length (mm.)	Outside diameter (mm.)	Minimum wall thickness of bottom (mm.)	Base diameter (mm.)	Length of taper from tip (mm.)	Outside diameter of open end (mm.)
6.5	118 ± 6	17.25 max.	----	4.0 ± 0.3*	----------	----------
15.0	118 ± 2	16.9 ± 0.35	0.51	6.5 ± 0.5	42.0 ± 5.0	19.25 ± 1.0
50.0	132.5 ± 2.5	28.0 ± 1.0	----	---------	----------	----------
100.0	See ASTM D 91					

*Inner diameter

3.2.2.3 Graduation markings. Graduation markings shall be narrow, sharply defined etched or engraved lines filled with a permanent pigment; etched through a vertical band fired into the glass; or by application of an enamel fired onto the color band without etching; or by application of a stain which is fired into the glass without etching; or by application of enamel fired onto the glass without etching. Graduation markings shall show no appreciable change when viewed with the naked eye after permanency of color tests as specified in 4.4.1 except for the loss of gloss.

3.2.2.3.1 Capacity 6.5 ml. The 6.5 ml. capacity centrifuge tube shall be marked at each 0.01 ml. from 0.01 ml. to 0.4 ml.; each 0.1 ml. from 0.4 ml. to 1.0 ml.; 1.5 ml.; each 1.0 ml. from 2.0 ml. to 5.0 ml.; and 6.5 ml. the main graduations shall be numbered and are 0.1, 0.2, 0.3, 0.4, 1.0, 2.0, 3.0, 4.0, 5.0, and 6.5. Intermediate graduations shall be approximately one half the length of the main graduations; shall not be numbered; and are 0.05, 0.15, 0.25, 0.35, 0.5, 0.6, 0.7, 0.8, and 0.9. The remaining graduations shall be lesser graduations shorter in length than the main or intermediate graduations and shall not be numbered. The graduations shall be accurate within the following tolerances:

From 0 ml. to 0.4 ml. ± 0.01 ml.
From 0.4 ml. to 1.0 ml. ± 0.05 ml.
From 1.0 ml. to 6.5 ml. ± 0.20 ml.

3.2.2.3.2 Capacity 15 ml. The 15 ml. capacity centrifuge tube shall be marked at each 0.1 ml. graduation from 0.1 ml. to 15 ml. The main graduations shall be at whole ml. values from 1 to 15. Intermediate graduations shall be at one-half ml. values from 0.5 ml. to 14.5 ml. Graduations of a shorter length shall be at one-tenth ml. values from 0.1 ml. to 14.9 ml. All main and intermediate graduations shall be at least 7 mm. long with the lengths in each category held to ± 0.5 mm. and all the shorter graduations shall be of equal length ± 0.5 mm., not shorter than 3 mm. long. The graduations shall be accurate within the following tolerances:

From 0 to 0.2 ml. ± 0.05 ml.
From 0.2 ml. to 1 ml. ± 0.1 ml.
From 1 ml. to 3 ml. ± 0.15 ml.
From 3 ml. to 15 ml. ± 0.20 ml.

The main graduations shall have and the subdivisions below 1 ml. may have Arabic numerals on projections of the graduations. All other graduations shall not be numbered. The graduations shall be perpendicular to the long axis of the tube and shall be parallel to each other. An imaginary line bisecting each graduation shall be straight. The numbers shall be in a straight line perpendicular to the long axis of the tube and shall be parallel to each other. The top of the graduated scale shall be at least 10 millimeters from the open end of the tube.

3.2.2.3.3 Capacity 50 ml. The 50 ml. capacity centrifuge tube shall be marked at each 0.5 ml. from 0.5 ml. to 10 ml. and each 1 ml. from 10 ml. to 50 ml. The main graduations shall be numbered and are 1, 2, 4, 6, 8, 10, 20, 30, 40, and 50 ml. Intermediate graduations shall be the same length as the main graduations, shall not be numbered, and are 3, 5, 7, 9, 15, 25, 35, and 45 ml. The remaining graduations are lesser graduations shorter in length than the main or intermediate graduations and shall not be numbered.

3.2.3 Type II.

3.2.3.1 Shape.

3.2.3.1.1 Conical bottom. All of the tubes except one of the 50 ml. capacity tubes shall have conical bottoms. The tip of the conical bottom shall be convex and thickened.

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3.2.3.1.2 Round bottom. One of the 50 ml. capacity tubes shall have a uniformly rounded bottom. The capacity shall be measured 9 mm. from the open end.

3.2.3.1.3 Pourout lip. The 50 ml. capacity tubes shall have a well formed pourout lip. The pourout lip shall pour envenly when tested as specified in 4.4.2.

3.2.3.1.4 Screw tube. One of the 50.0 ml. capacity tubes shall have a neck with GCMI 24-410 1/ threads.

1/Glass Containers Manufacturers Institute.

3.2.3.1.4.1 Screw-cap. The screw-cap shall be made of a plastic suitable for autoclaving, and shall have a liner of rubber glued in.

3.2.3.2 Dimensions. Dimensions of type II centrifuge tubes shall be as shown in table II.

3.3 Resistance to centrifugal force. Tubes shall not break or crack when tested as specified in 4.4.3.

3.4 Thermal shock. Centrifuge tubes shall not crack or break when tested as specified in 4.4.4.

3.5 Identification marking. Each centrifuge tube shall be permanently and legibly marked with the manufacturer's name or trademark. Each centrifuge tube shall have an area suitably located for laboratory marking.

3.6 Workmanship. Centrifuge tubes shall be symmetrical in shape, uniform in thickness except at the tip of the bottom, and free from defects that detract from their appearance or impair their serviceability.

TABLE II. Dimensions of type II centrifuge tubes

Capacity (ml.)	Length (mm.)	Outside diameter (mm.)	Minimum wall thickness of bottom (mm.)	Base diameter (mm.)	Length of taper from tip (mm.)	Outside diameter of open end (mm.)
0.5	58 ± 2	6 ± 0.25	----	--------	----------	----------
1.0	61 ± 2	8.25 ± 0.25	----	--------	----------	----------
2.0	66 ± 2	10.75 ± 0.25	----	--------	----------	----------
3.0	74 ± 2	10.75 ± 0.25	----	--------	----------	----------
5.0	101 ± 2	13.0 ± 0.5	----	--------	----------	----------
12.0	117.5 ± 2.5	17.0 ± 1.0	----	--------	----------	----------
15.0	118 ± 2	16.9 ± 0.35	0.51	6.5 ± 0.5	42 ± 5.0	19.25 ± 1.0
40.0	132.5 ± 2.5	28.0 ± 1.0	----	--------	----------	----------
50.0 2/	132.5 ± 2.5	28.0 ± 1.0	----	--------	----------	----------
50.0 3/	151.0 ± 2.5	28.0 ± 1.0	----	--------	----------	----------
50.0 4/	116 ± 2	28.0 ± 1.0	0.61	--------	----------	34 max. including pourout lip

2/conical bottom
3/conical bottom with screw-cap
4/round bottom

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of chemical properties) of the item.

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product shall be the tube.

Characteristic	Requirement	Test procedure	Inspection level
Permanency of marking	3.2.2.3	4.4.1	S-1
Pour	3.2.3.1.3	4.4.2	S-1
Resistance to centrifugal force	3.3	4.4.3	S-2
Resistance to thermal shock	3.4	4.4.4	S-2

4.3 Examination. The tube shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

Categories	Defects*
Major:	
101	Tube not free of sharp edges.
102	Tube not free of chipping.
103	Tube not free of cracks.
104	Tube not free of protrusions.
105	Graduation marking not complete, correct, legible, or correctly positioned.
Minor:	
201	Tube not free of striae.
202	Tube not free of bubbles.
203	Tube not free of seeds.
204	Identification marking not complete, correct, or legible.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Examination for preparation for delivery requirements. An examination shall be made to determine whether packaging, packing and marking complies with section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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TABLE III. Preparation for delivery defects

Levels A and B preparation for delivery (section 5)

Categories	Defects
Critical:	None defined.
Major:	
101	Unit, intermediate, or shipping container not as specified.
102	Unlike centrifuge tubes packaged per unit, intermediate, or shipping container.
103	Closure of unit, intermediate, or shipping container not as specified.
104	Unit, intermediate, or shipping container damaged.
105	Cushioning material or fiberboard fittings not as specified.
106	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Permanency of color in graduation markings. A different tube shall be used for each test specified below.

4.4.1.1 Resistance to alkali. The centrifuge tube shall be completely immersed in 1N sodium hydroxide, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The tube shall then be removed, thoroughly rinsed with distilled water and,dried by rubbing vigorously with a cloth.

4.4.1.2 Resistance to acid.

4.4.1.2.1 Hydrochloric acid. The centrifuge tube shall be immersed completely in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The centrifuge tube shall then be removed from the acid, thoroughly rinsed with water, and dried by rubbing vigorously with a cloth.

4.4.1.2.2 Sulfuric acid. The centrifuge tube shall be immersed completely in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The centrifuge tube shall then be removed, thoroughly rinsed in water,and dried by rubbing vigorously with a cloth.

4.4.2 Pour test. Fill the tube to approximately 90 percent of capacity with water. Pour approximately one-fourth of the water from the tube at such a rate that the water flows in a continuous stream and falls free from the lip of the tube. If the stream of water clings to the outside of the tube and flows off the bottom, it shall be considered cause for rejection.

4.4.3 Resistance to centrifugal force.

4.4.3.1 Centrifuge. The centrifuge may be any type or model commercially available or normally employed in chemical laboratories and capable of speeds in excess of 2750 revolutions per minute (r.p.m.). The head of the centrifuge shall be such that the tips of opposite centrifuge tubes shall be 10 to 11 inches apart when the centrifuge is in operation.

NOTE: Where a centrifuge of 10 to 11 inch diameter swing is not available, a centrifuge with larger or smaller diameter swings may be used. The centrifuge speed, in r.p.m., will require adjustment according to the following equation:

$$r.p.m. = 265 \sqrt{\frac{rcf}{d}}$$

Where:

r.p.m. = centrifuge speed in revolutions per minute
rcf = relative centrifugal force = 1125 ± 50
d = diameter of swing (from tip to tip of opposite centrifuge tubes)

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4.4.3.2 Procedure. The centrifuge tubes shall be filled with a quantity of water equal to the capacity as shown in tables I and II for each respective type and capacity of centrifuge tube tested. The centrifuge tubes for test shall then be placed in the centrifuge in balanced positions and brought to a speed of 2750 r.p.m. for 30 minutes. The centrifuge shall then be shu shut off and allowed to stop without use of the hand brake.

4.4.4 Thermal Shock. Centrifuge tubes shall be tested in accordance with Fed. Test Method Std. No. 626/401, using an oven temperature of 180° C.

4.5 Acceptance/rejection criteria. If any centrifuge tube fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The number of centrifuge tubes specified (see 6.2), of the same type and capacity, shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636 type CF, class domestic; or PPP-B- 676, type, class, and style optional. The box shall be provided with fiberboard fittings or shall contain sufficient cushioning material conforming to PPP-C-843, type II, class B, to prevent shifting of or damage to the contents. Box closure shall be secured with tape conforming to PPP-T-45.

5.1.1.2 Intermediate packaging. The number of unti packages specified, (see 6.2) shall be intermediately packaged in a snug-fitting box conforming to PPP-B-636, type CF, class domestic. The box shall be closed in accordance with the appendix of the box specification. All centrifuge tubes in an intermediate package shall be of the same type and capacity.

5.1.2 Level C. Centrifuge tubes shall be packaged in accordance with the manufacturer's commercial pratice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Centrifuge tubes shall be packed in a box conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and shall contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 Level B. The number of centrifuge tubes specified (see 6.2), packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box conforming to PPP-B-636, class domestic, type CF.

5.2.3 Level C. Centrifuge tubes shall be packed to insure carrier acceptance and safe delivery at first destination for immediate use. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addtion to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Centrifuge tubes covered by this specification are intended for general use. In addition, the following are designed for usage with particular tests.

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6.1.1 Type I, capacity 6.5 ml. This centrifuge tube is intended for use for the determination of protein in cerebrospinal fluid by the McNaught method and protein in urine by the methods of MacKay, Shevky, and Stafford.

6.1.2 Type I, capacity 100 ml. This centrifuge tube is intended for use for the determination of the precipitation number of lubrication oils by ASTM D 91 and the determination of water and sediment in crude mineral oils and fuel oils by ASTM D 96.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Type and capacity required (see 1.2.1).
(d) Number of centrifuge tubes per unit, intermediate, and shipping container required.
(e) Special markings required (see 5.3.1 and 5.3.2).

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard and military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Supersession data. This specification supersedes Military Specifications MIL-T-3249, July 12, 1960 and MIL-T-16369 (BuMed), June 1951. Centrifuge tubes of superseded military specifications are covered by this specification as follows:

TABLE III. Superseded military specifications

NNN-T-780B Centrifuge Tube	Superseded specification Centrifuge Tube
Type I, 15 ml.	MIL-T-16369 (BuMed), 15 ml., type I
Type I, 100 ml.	MIL-T-3249 (1 centrifuge tube, only)
Type II, 15 ml.	MIL-T-16369 (BuMed), 15 ml., type II
Type II, 50 ml. round bottom	MIL-T-16360 (BuMed), type III

MILITARY INTEREST:

Custodians:

Army - MU
Navy - MS

Review activities:

Army - MD
Navy - MS
DSA - DP

User activities:

Army - MI, SM

Preparing activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

AGR
COM-NBS
GSA-FSS
HEW
VA-DMS

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, **and mail to: Gen. Services Admin., FSS (FMSC), Washington, D. C. 20406.**

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

NNN-T-780B, TUBE, CENTRIFUGE, GLASS

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

FORM GSA FEB 69 2200

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